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Common Myofascial Interventions

Myofascial interventions are a group of devices used to provide tissue massage to the muscles, fascia, and soft tissues of the body. These interventions have become popular among professionals and consumers, stimulating the creation of different products and emerging research. There are several myofascial interventions such as myofascial rolling (e.g., foam roller) (MR), instrument assisted soft-tissue massage (IASTM), mechanical percussion (MP), flossing bands (FB), kinesiology tape (KT), cupping, and therapeutic massage. These interventions are often combined with stretching for pre-event, post-event, rehabilitation, and therapeutic purposes (Lee et al., 2018; Ortiz et al., 2019; Poppendieck et al., 2016; Su et al., 2017).

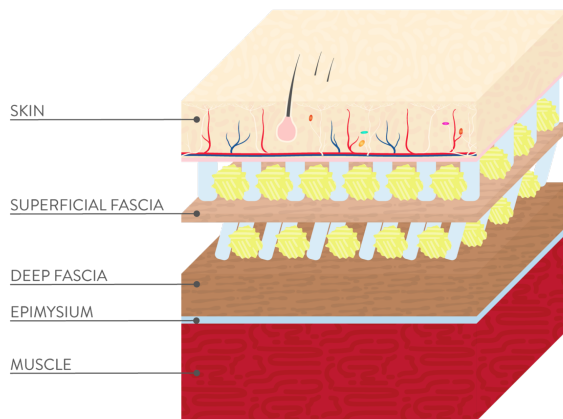


Figure 1: Layers of the Myofascial System

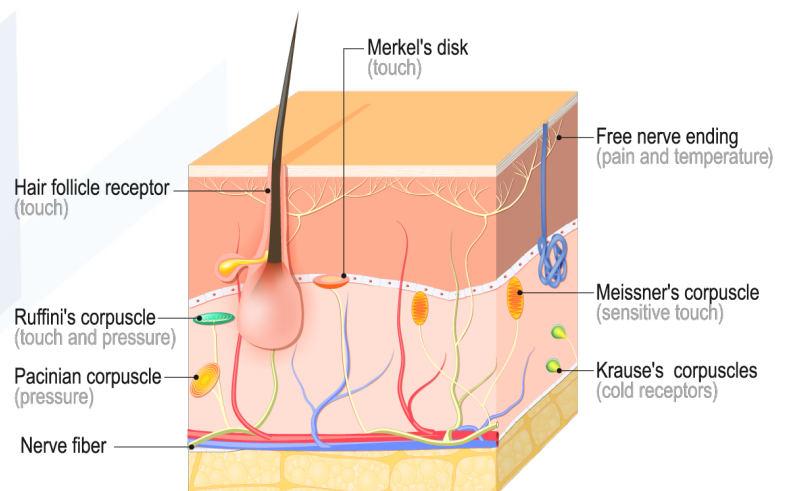
glands, and nerves. It is the largest organ in the body and contains three layers: epidermis, dermis, and hypodermis (Wong et al., 2016). Just below the deepest skin layer is the superficial fascial layer, deep fascial layer, muscles, tendons, ligaments, and joint capsules (Benjamin, 2009). Within all these layers is a complex network of cellular structures, vessels, and nerves (**Figure 1: Layers of the Myofascial System**).

Perhaps, the most relevant structures in the myofascial system are the different afferent (sensory) nerve receptors that span through the different layers of the skin, fascia, joint capsule, ligaments, tendons, and muscles. Most notable are the mechanoreceptors, which respond to outside forces, such as touch, pressure, stretching, sound waves, and motion, and transmit impulses through sensory nerves, which, in turn, enable humans to detect touch, sounds, and the motion of the body and to monitor the position of muscles, bones, and joints (proprioception) (**Figure 2: Mechanoreceptors**) (García-Piqueras et al., 2019; Iheanacho & Vellipuram, 2020). The muscle spindle and Golgi tendon organ are two other notable

In this course, these myofascial interventions will be introduced and described in more detail, including common ways they are used by health and fitness professionals and clients. The science behind these devices will be discussed along with common precautions and contraindications. The integration of these interventions in a corrective exercise program that includes stretching will also be discussed.

Myofascial System: Review

The fitness professional needs to have a good understanding of the skin and myofascial system to fully understand the science behind these myofascial interventions. The myofascial system contains different layers composed of the integumentary system, fascia, and muscle. The integumentary system consists of the skin, hair, nails,



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Figure 2: Mechanoreceptors

receptors that help to regulate muscle contractions and human movement. Free nerve endings also span through the different skin layers, providing feedback for temperature and pain (Matthews, 2015).

The stimulation of afferent nerves through the compression or distraction of the myofascial system induces an immediate efferent response by the central nervous system. Researchers have studied this connection when using different myofascial interventions, leading to two current scientific theories, which will be discussed in the next section.

Myofascial Rolling

Myofascial rolling (MR) is perhaps the most popular intervention among professionals, emerging as one of the top 20 fitness trends over the past 5 years (Thompson, 2015, 2016). The existing body of research for myofascial rolling has grown, with many researchers finding positive outcomes using these devices as part of a warm-up and recovery from strenuous exercise. Furthermore, MR has also been shown to have therapeutic benefits, such as modulating pain and increasing joint range of motion (Schroeder & Best, 2015; Wilke et al., 2020). Fitness professionals and clients often use these rollers as part of the pre-exercise warm-up, between exercises, or during the post-exercise cool-down (Cheatham, 2019).

MR devices often include foam rollers, roller sticks, and roller balls that are available in many shapes, sizes, materials, surface textures, and densities (**Figure 3: Example MR Devices**). *Foam rollers* are typically available in two sizes: standard (6-inch x 36-inch) and half size (6-inch x 18 inches) (Cheatham & Stull, 2018). With foam rolling, the client uses their bodyweight to apply pressure to the soft tissues during the rolling motion. For example, the individual may lay prone on the roller (e.g., plank positions) with a roller under their quadriceps then roll back and forth along the length of the muscle for a specified amount of time.



Figure 3: Example MR Devices

Roller massage sticks often include two handles connected by a solid plastic cylinder with a plastic or dense foam outer covering (Bradbury-Squires et al., 2015; Halperin et al., 2014; Sullivan et al., 2013). The individual grasps the handles and applies a compressive force while moving the bar in different directions across the tissues. The pressure during the rolling action is determined by the force from the upper extremities. *Roller massage balls* also come in many shapes, densities, and sizes (Cheatham et al., 2020). Due to the smaller diameter, individuals often use the ball to provide a deeper massage on smaller areas of the body. For example, roller balls are often used to roll the bottom of the foot. The individual may be sitting or standing and put their foot on top of the ball and roll in different directions. The pressure can be changed by pushing down onto the ball.

The most common methods are for the client to lay on a foam roller or roller ball to massage a specific region of the body. However, these methods may not reflect the existing research, which needs to be considered by the professional when adopting these approaches. NASM recommends the use of MR as part of the Corrective Exercise Continuum,

which is an evidence-based strategy to correct postural and movement dysfunctions among clients. NASM further recommends using MR as a flexibility modality within each phase of the Optimum Performance Training® (OPT™) model.

Mechanical Percussion

Mechanical percussion has also become a popular intervention among fitness, sports, and healthcare professionals (**Figure 4: Mechanical Percussion**). Mechanical percussion devices are electric- or battery-powered and use a shaped tip to provide a rapid compression force to the myofascia. The action of these devices is similar to manual tapotement or



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Figure 4: Mechanical Percussion

percussion, which is a specific technique in therapeutic massage where a clinician applies a rapid, compressive striking of the myofascia with the hands (Behm et al., 2013). Mechanical percussion devices may produce a greater effect by impacting the myofascia at a higher rate of frequency and force.

There are many manufactured models on the market with different speeds and application tips. The current body of evidence is sparse with no current published studies on these devices. A survey study conducted by the author revealed health and fitness professionals use these devices for pre- and post-exercise, to decrease pain, and to enhance myofascial mobility (Cheatham, in press). However, due to the lack of research, professionals are using their own preferred methods to administer these devices to clients.

CAUTION!

Mechanical percussion devices are the newest and most popular myofascial devices available to health and fitness professionals and clients. Unfortunately, there is currently no current research on this technology, which leaves a gap in understanding the clinical application of these devices. The fitness professional should always consider the safety and legal risks of using such devices on clients.



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Figure 5: IASTM

Instrument Assisted Soft-Tissue Mobilization

Instrument Assisted Soft-Tissue Mobilization (IASTM) is a popular tool intervention applied by healthcare professionals (**Figure 5: IASTM**). The treatment includes the use of specially designed instruments to provide a massage to the myofascia (Cheatham et al., 2019). IASTM tools come in many shapes, sizes, textures, and materials (e.g., stainless steel). The use of the instrument is thought to provide a mechanical advantage for the clinician by allowing deeper tissue penetration, vibration feedback sense, and more specific treatment, while also reducing imposed stress on the professional's hands (Cheatham et al., 2019). There are various IASTM companies that teach their own treatment approach and design their

own instruments (e.g., specific material or instrument shape). The existing body of research on these tools is growing and supports the use of these tools for treatment of different musculoskeletal conditions, reducing pain, and increasing joint range of motion (Cheatham et al., 2019; Cheatham et al., 2016; Seffrin et al., 2019).

CAUTION!

Fitness professionals should refer clients to professionals who have the proper credentials and training to complete this therapy. This technique can cause tissue damage if used improperly. For example, the misuse of IASTM can cause red skin blemishes and bruising, which can be a safety risk for the client.

IASTM is used by trained professionals such as chiropractors, physical therapists, and certified athletic trainers. The use of IASTM tools to provide a deep massage to the myofascia requires a certain level of knowledge and training. IASTM is not commonly used among fitness professionals who do not have the proper training and licensure.

Floss Bands

Myofascial flossing, otherwise termed as *voodoo flossing* or simply as *flossing*, is another emerging myofascial intervention that includes wrapping a band around a body part to create myofascial compression. Flossing research continues to emerge supporting the use of the bands for recovery after strenuous exercises, increasing joint ROM and enhancing performance (Driller et al., 2017; Driller & Overmayer, 2017; Mills et al., 2019; Prill et al., 2019). Several manufacturers make floss bands, which are typically made from latex or non-latex and are available in different widths, band thickness, and lengths (**Figure 6: Floss Band**).

Myofascial flossing is performed by teaching the client to wrap the band around a body part using a 50% overlap pattern and using a stretch force of 50 to 90% (Cheatham & Baker, 2019; Kiefer et al., 2017). Once the band is applied, the client then performs up to 2 minutes of active or passive movements to induce tissue flossing (Borda & Selhorst, 2017; Cheatham & Baker, 2019). Myofascial flossing creates a tangential shearing or wringing-towel effect by creating compression around the muscle groups; then, the movements are intended to help mobilize the myofascia.

Health and fitness professionals and clients seem to be using floss bands for pre-exercise, between exercises, and for the post-exercise cool-down. Floss bands are popular among the CrossFit community, where different applications are shared among enthusiasts. Thus far, the bands appear to be both safe and provide positive benefits.



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Figure 6: Floss Band

CAUTION!

Fitness professionals should be cautious when using these bands on clients with medical conditions or latex allergies. The fitness professional should not confuse these bands with blood flow restriction, which is a completely different treatment and science. This technique should also not be confused for nerve flossing, which is a neurological treatment administered by licensed healthcare professionals.

Kinesiology Tape

Kinesiology tape (KT) is another popular intervention widely used by health and fitness professionals and consumers. The tape has elastic properties and comes in different sizes, widths, material, and textures (**Figure 7: Kinesiology Tape**). KT is often applied for injury prevention, performance, and rehabilitation. KT has a large body of research on the therapeutic effects; however, most study outcomes are variable with different researchers reporting different outcomes



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Figure 7: Kinesiology Tape

Cupping

Cupping has been a long-standing myofascial intervention practiced for thousands of years from Egypt to Greece to China. Different civilizations have used cupping as part of their medical practice (Soliman et al., 2018). Historically, cupping has been used to treat different diseases, musculoskeletal conditions, skin conditions, and other medical ailments (Mehta & Dhapte, 2015; Soliman et al., 2018). In the United States, cupping has become a popular complementary and alternative myofascial intervention over the past 10 years. There are two main types of cupping: dry and wet.

Dry cupping uses silicone, plastic, or glass cups to distract the skin and connective tissue into the cups by creating a negative pressure that suctions the tissues (**Figure 8: Dry Cupping**).

With the silicone cups, the professional presses the cup into the skin, which pushes out the air from inside the cup, and then releases the cup, causing negative pressure inside the cup, which suctions the tissues into the cup.

Many plastic or glass cups come with a one-way valve at the top of the cup in which a pneumatic pump can be used to take the air out, creating negative pressure. In the more traditional form of cupping, practitioners would use glass cups and light a cotton swab on fire and place it into the cup, producing a negative pressure, and then quickly put the cup on the skin, which would then suction the tissues. This technique has been used less frequently due to the risk of burns (Seifman et al., 2017).

Wet cupping uses the same suction component plus the addition of bloodletting, which is the process of puncturing the skin with a sterile sharp edge (e.g., razor blade) and allowing the “old” blood to leave the body. This process is supposed to have

from the tape (Montalvo et al., 2014). Several researchers attribute the mixed outcomes due to different study methods and the use of different tape brands that are manufactured differently (Boonkerd & Limroongreungrat, 2016; Matheus et al., 2017; Selva et al., 2019).

Professionals often use the 2-inch (5.08-cm) width tape and stretch the tape between 25 to 75% of the resting length when applying to a client. For example, a 25% to 50% stretch has been recommended for treatment of fascia, circulatory, stimulating, and inhibiting muscle activity (de Jesus et al., 2017; Montalvo et al., 2014; Reneker et al., 2018). A stretch between 25 to 75% may provide optimal therapeutic effects for treatment of musculoskeletal conditions and sports performance (de Jesus et al., 2017; Montalvo et al., 2014; Reneker et al., 2018).

CAUTION!

With the use of kinesiology tape, the fitness professional needs to consider if using KT is within their scope of practice. Fitness professionals should have a comprehensive understanding of how to apply the tape, indications, contraindications, and current research. This can be done with formal training.



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Figure 8: Dry Cupping

CAUTION!

The fitness professional needs to understand the clinical guidelines for cupping intervention because clients may undergo this treatment. For example, both methods may create round red blemishes (e.g., shape of cup) to the skin that may last for days. The blemishes are an acceptable side effect of treatment but do not indicate extensive tissue damage.

therapeutic benefits for the individual by stimulating various physiological responses (Soliman et al., 2018). There are obvious legal, safety, and hygiene risks with wet cupping. Most states in the United States do not allow this treatment unless the professional is properly licensed. The use of cupping by fitness professionals is uncommon unless they have the **proper training and licensing**. Dry cupping is the most common type in the United States and is mainly used to treat musculoskeletal conditions and promote recovery after strenuous exercise.

Therapeutic Massage

As a complimentary intervention to their fitness program, clients may choose to get therapeutic massage to address myofascial dysfunctions as a supportive intervention alongside their fitness program. A popular form of therapeutic massage is manual myofascial release, which is a soft-tissue technique that can be either direct or indirect and is frequently combined. As a massage technique, it utilizes the principles of biomechanical loading of soft tissue and the neural reflex changes by stimulation of mechanoreceptors in the fascia (Schmidt & Willis, 2007). Manual myofascial release is also used to treat a variety of chronic disorders in which the muscle tissue is stretched and manipulated to relieve tension in the myofascia (O'Toole, 2016).

The current evidence on manual myofascial release supports its efficacy to produce positive physiological (Cao et al., 2013; Cao et al., 2015; Guo et al., 2017) and therapeutic benefits (Ajimsha et al., 2015; McKenney et al., 2013). Thus, therapeutic massage may produce the same effects as the other interventions, but it is administered by a trained professional, just like IASTM and cupping. The other myofascial interventions may be performed by the client, which needs to be considered by the fitness professional.

CAUTION!

Manual myofascial release is a specialized form of therapeutic massage and is administered by a licensed healthcare professional. The fitness professional must understand the clinical guidelines for therapeutic massage because clients may get a massage as a complimentary treatment.

Myofascial Interventions: Scientific Theories

For the past 10 years, many of these interventions have been classified as a type of self-myofascial release (SMR) or myofascial release (MR). These terms may not clearly represent what is physiologically occurring during or after the intervention. The term *release* alludes to a “setting free” (O'Toole, 2016) of the myofascial tissues, which is contrary to the growing body of research that suggests that a more comprehensive physiological response occurs in the body from these interventions (Beardsley & Škarabot, 2015; Cheatham et al., 2015; Hotfiel et al., 2017).

All the myofascial interventions discussed in this course, excluding cupping, directly compress or stretch the tissues, which may produce local and global effects. Cupping does create a stretching and decompressing effect by pulling the myofascial layers away from the body. Researchers posit that cupping may induce similar responses as the other interventions (Aboushanab & AlSanad, 2018).

The growth in myofascial research has led researchers to propose two scientific theories: mechanical and neurophysiological effect (**Figure 9: Myofascial Intervention Theories**). For the *mechanical effect*, the pressure of the device or wrap may change the viscoelastic properties of the local myofascia by mechanisms such as thixotropy (reduced viscosity), reducing myofascial restriction, fluid changes, and cellular responses (Jay et al., 2014; Kelly & Beardsley, 2016). For the professional, these changes may be observed as a greater lengthening or stretch tolerance of the muscle and surrounding tissues as measured by changes in joint ROM.

For the *neurophysiological effect*, the mechanical pressure from the device or wrap may influence tissue relaxation and pain reduction in the target and surrounding tissues through central nervous system activity, producing a more global effect (Aboodarda et al., 2015; Grabow et al., 2019; Young et al., 2018). For *tissue relaxation*, the local pressure from the devices or wrap may affect the local and surrounding myofascia through the mechanical effects noted in the prior

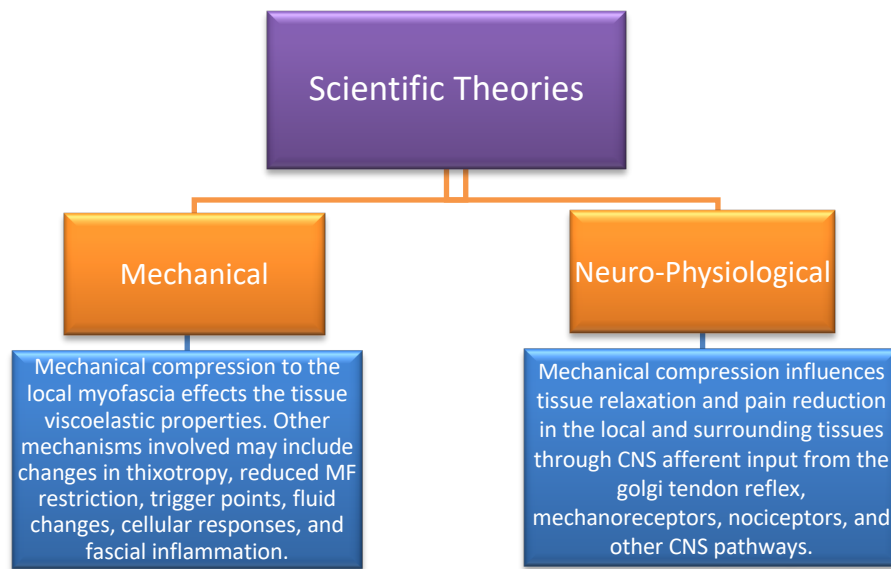


Figure 9: Myofascial Intervention Theories

section. The local compression may also induce local blood flow changes (Hotfiel et al., 2017; Okamoto et al., 2014). Researchers have postulated that the pressure from devices may modulate pain through stimulation of skin receptors (Aboodarda et al., 2015), mechanoreceptors (Young et al., 2018), and other central nervous system pathways (Aboodarda et al., 2015; Cavanaugh et al., 2017; Sullivan et al., 2013).

CHECK IT OUT!

By applying the theories in Figure 9, the professional will be able to observe post-intervention changes. For the mechanical effect, the professionals may observe a greater lengthening or stretch tolerance of the muscle and surrounding tissues as measured by changes in joint ROM. For the neurophysiological effect, the professional may observe global tissue effects through improved kinetic chain movement efficiency such as improvements in the overhead squat. Global pain modulation can be observed in the local and surrounding tissues via some type of reciprocal inhibition and crossover phenomenon. For example, rolling the quadriceps can modulate pain in the ipsilateral hamstring (reciprocal inhibition) and the opposite quadriceps (crossover phenomenon).

For the professional, the global tissue effects can be observed through improved kinetic chain movement efficiency such as improvements in the overhead squat (Monteiro et al., 2017). The global pain modulation can be observed in the local tissues being treated (Wilke et al., 2018) and it surrounds tissues via some type of reciprocal inhibition and crossover phenomenon (Cheatham & Kolber, 2018). The two current scientific theories suggest that more complex local and global responses occur versus a more simplistic theory of myofascial release. These complex responses may be triggered by low, moderate, or high pressure on the tissues, lending evidence to the sensitivity of the neuromyofascial system to external forces (Grabow et al., 2017). The fitness professional should consider that both of these theories support responses that may occur at the same time.

Corrective Exercises: Myofascial Interventions and Stretching

The corrective exercise prescription for clients with postural dysfunction, flexibility, and mobility deficits may involve a combination of corrective strategies that include myofascial interventions and stretching (Su et al., 2017). These interventions can be done in isolation or as a multimodal program, depending on the client's needs and health status. NASM recommends the Corrective Exercise Continuum that includes four phases: inhibit, lengthen, activate, and integrate. The first two phases often include a combination of myofascial rolling and different types of stretching such as

CAUTION!

It is not uncommon to see professionals share their methods of application for myofascial interventions and stretching on social media and/or Internet videos. The fitness professional should consider that these shared methods may be from personal preference and not evidence-based sources. This should be considered before integrating such techniques into a client's program. The professional scope of practice should also be considered when utilizing different myofascial interventions.

static and neuromuscular. MR is the most accessible intervention for the fitness professional and falls within the scope of practice. Some of the other interventions may or may not be used by fitness professionals.

The research suggests that a multimodal program consisting of myofascial rolling, static stretching, and dynamic movements may produce favorable outcomes versus a single intervention (Behara & Jacobson, 2017; Su et al., 2017). Fitness professionals may often combine these interventions as part of a warm-up or post-exercise cool

down. For example, short bouts (less than 30 seconds per muscle group) of myofascial rolling for the major muscle groups followed by dynamic movements may help to prep the body for physical activity. Or longer bouts of rolling (greater than 60 seconds) followed by static stretching may help the lengthen and relax the myofascial tissues after strenuous exercise. The combination of interventions can fit into most client programs and should be considered within their exercise prescription.

General Precautions and Contraindications

The myofascial interventions detailed in this course all have common and unique medical precautions and contraindications. This section will discuss common precautions and contraindications among all interventions. Medical precautions are conditions that may be unsafe for clients undergoing a myofascial intervention.

General medical precautions for most myofascial interventions include but are not limited to hypertension, osteopenia, pregnancy, diabetes, varicose veins, bony prominences or regions, abnormal sensations (e.g., numbness), sensitivity to pressure, inability to position body or perform the intervention, young children, older individuals, and scoliosis or spinal deformity (Cheatham et al., 2019; Cheatham & Stull, 2018). General medical contraindications are presented in **Table 1**.

Table 1: Myofascial Intervention General Contraindications

- | | |
|---|--|
| <ul style="list-style-type: none"> • Skin rash, open wounds, blisters, local tissue inflammation, bruises, or tumors • Osteoporosis • Bone fracture or Myositis Ossificans • Acute or severe cardiac, liver, or kidney disease • Neurologic conditions resulting in loss or altered sensation (e.g., multiple sclerosis) • Systemic conditions (e.g., diabetes) • Connective tissue disorders (Marfan syndrome) • Medications that thin blood or alter sensations • Chronic pain conditions (e.g., rheumatoid arthritis) • Pregnancy (consult MD/OBGYN) • Extreme discomfort felt by patient | <ul style="list-style-type: none"> • Deep Vein Thrombosis • Cancer or malignancy • Hypertension • Acute infection (viral or bacterial), fever, or contagious condition • Bleeding disorders (hemophilia) • Recent surgery or acute injury • Peripheral vascular insufficiency or disease • Direct pressure over varicose veins • Direct pressure over face, eyes, arteries, veins, or nerves • Direct pressure over bony prominences or regions (e.g., lumbar vertebrae) • Severe scoliosis or spinal deformity |
|---|--|

Note: The general medical precautions and contraindications listed in this section are not all inclusive and should be considered a starting point for professionals to build their own list based on their client population.

Note that some conditions such as, but not limited to, pregnancy, diabetes, varicose veins, and hypertension can be considered either precautionary or contraindicated depending on the client (Ernst, 2003; Ranjbaran et al., 2017; Yin et al., 2014). These conditions and others are listed in both categories.

Equipment Cleaning Recommendations

Many of the myofascial interventions discussed in this course can be found in different health and fitness settings. Often, these tools are shared among individuals. Regular cleaning of the objects is advised to prevent the spread of germs (bacteria, viruses, and fungi). Disinfecting the device surfaces often includes the use of an EPA approved commercial cleaner in the form of a spray or a ready-to-use wipe. The CDC and EPA recommend using at minimum an approved low-level chemical disinfectant or more preferably an intermediate-level disinfectant (**Table 2: CDC Levels of Disinfection**) (Nielsen et al., 2012; Veiga-Malta, 2016). Common intermediate-level disinfectants include 70% isopropyl alcohol, Lysol® brand disinfectant products, and Clorox® brand disinfectant products. Bleach can also be used to clean (e.g., 5 tablespoons or 1/3 cup bleach to 1 gallon of water) but can be corrosive to some surfaces.

Table 2. CDC Levels of Disinfection	
Intermediate-level disinfection	These disinfectants kill mycobacterium, most viruses, and bacteria with a chemical germicide registered as a "tuberculocidal" by EPA.
Low-level disinfection	These disinfectants kill some viruses and bacteria such as HIV with a chemical germicide registered as a hospital disinfectant by the EPA.

Fitness professionals should follow the directions when using a commercial disinfectant. Most products require the liquid to remain on the surface for a specific amount of time to be effective. This is based upon the product's "wet time" (Rutala & Weber, 2016). For example, to sanitize a surface, the Lysol® brand wipes recommend a "wet time" of 10 seconds and to disinfect a "wet time" of 4 minutes. Each product will have its own recommended time that is specified on the label.

Disinfecting Procedures

When disinfecting myofascial devices, the fitness professional is advised to follow five essential steps in the cleaning process. First, wash hands before cleaning. Wash hands with soap and water or rub hands together using an alcohol-based hand sanitizer (e.g., 60-70% alcohol) for a minimum of 20 seconds (Fuls et al., 2008; Honda & Iwata, 2016). Second, wear personal protective equipment (PPE) (e.g. gloves, mask) during cleaning, if needed (Honda & Iwata, 2016). Third, if using wipes, the surface should be wiped in the same direction to prevent contamination versus back and forth. Multiple wipes may need to be used. If using a spray, the surface can simply be sprayed down. Fourth, the disinfectant liquid must remain on the surface for a specific amount of time based upon the product recommendations. Fifth, when finished dispose of any PPE, cleaning materials, and re-wash hands (**Figure 10: Five Disinfection Steps**).

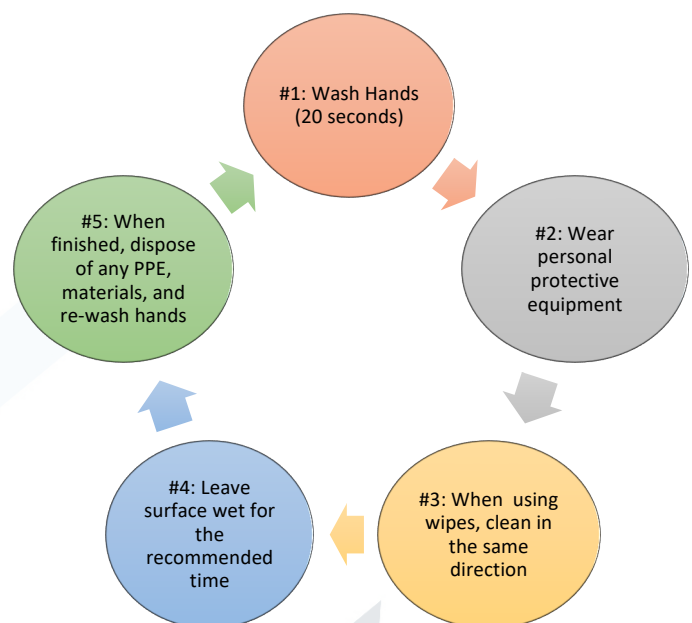


Figure 10: Five Disinfection Steps

Shared myofascial devices such as a foam roller may need to be disinfected several times a day. After disinfecting, the professional may want to flush the device with soap and clean water to wash away any dried chemical disinfectant, which could irritate a client's skin.

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